

Structural Limits of Adversarial Identity Scaling in Open Systems: An Axiomatic Resource-Cost Framework

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Abstract—Open and permissionless systems mitigate Sybil attacks by binding influence to scarce resources. However, *aggregate scarcity alone does not determine how the cost of sustaining multiple identities scales over time*. Many widely deployed resources—such as computation and financial capital—are *divisible and reusable*: they can be subdivided, transferred, and reapplied across identities and time without proportional marginal expenditure. As a result, adversarial effort may be amortized across identities, enabling large-scale identity replication at sublinear cost.

This paper develops an axiomatic framework for analyzing security-weighting resources through the lens of *adversarial cost scaling*. We identify a small set of resource properties—including divisibility, additivity of influence, temporal reusability, identity transferability, and throughput boundedness—that characterize the asymptotic growth of sustained adversarial cost.

Under a standard amortized coordination model, we establish an asymptotic boundary between two resource classes. Within the present abstraction, mechanisms grounded in divisible, reusable resources admit identity amortization, yielding $C(s, T) = o(sT)$; the marginal cost per additional identity does not grow with the time horizon. In contrast, throughput-bounded, non-transferable, window-local resources indicate a linear scaling regime, $C(s, T) = \Omega(sT)$, with marginal cost $\Delta(s, T) = \Omega(T)$.

Under the assumptions considered here, protocol rules that redistribute a fully divisible and reusable resource may be insufficient to enforce linear per-identity cost. The framework complements existing deployment-level and protocol-level analyses. Analytical illustrations accompany the theoretical results.

Index Terms—Sybil attacks, adversarial cost models, distributed security, permissionless systems, decentralized governance, identity scaling.

I. INTRODUCTION

Sybil resistance is a foundational requirement in open and permissionless systems. Existing approaches typically regulate adversarial influence by associating participation with scarce resources such as computational effort or financial stake. While such mechanisms are commonly analyzed in terms of aggregate adversarial resource ownership, many security-critical operations depend on the scaling behavior of identity-level participation cost.

This paper develops a structural framework for analyzing how the properties of a security resource determine the asymptotic structure of adversarial identity scaling.

A. Motivation

Open and permissionless systems must regulate participation under adversarial conditions in which identity creation is inexpensive and effectively unbounded. The classical formulation of this problem is the Sybil attack [1], where a single adversarial entity generates multiple identities to amplify influence within a distributed system. This challenge affects blockchain consensus, decentralized governance, reputation systems, committee selection, and distributed access-control mechanisms [2]–[10].

A common mitigation strategy binds influence to the possession or expenditure of a scarce resource. Proof-of-Work associates influence with computational effort [5], while Proof-of-Stake binds participation to financial capital [11]. Such mechanisms are typically analyzed in terms of aggregate resource ownership and adversarial thresholds.

However, many security-sensitive operations in open systems operate at the identity layer rather than the aggregate resource layer. Examples include governance participation, committee sampling, rate-limited interaction, and identity-scoped access control. In these settings, the relevant question is not only how much resource an adversary controls in aggregate, but how the cost of sustaining many identities scales with both the number of identities and the duration of participation.

A Resource-Structural Perspective. Many widely deployed security resources are divisible and reusable. Computational capacity and financial stake can often be partitioned, delegated, and reused across identities while preserving aggregate influence. Mining pools and stake delegation illustrate this property: once aggregate resource ownership is acquired, it may be redistributed across many identities without proportional renewal cost [9], [11]–[13].

This motivates a resource-centric perspective on Sybil resistance: rather than focusing exclusively on aggregate adversarial ownership, we study how the structural properties of a security resource determine the asymptotic scaling behavior of sustained identity cost.

What structural properties must a security resource satisfy for adversarial participation cost to grow

proportionally with the number of sustained identities?

B. Contribution and Framing

This paper develops a structural framework for analyzing adversarial identity scaling in open systems. The objective is not to replace deployment-specific security analyses, but to isolate how resource structure influences the asymptotic scaling of sustained participation cost.

On the nature of the contribution. The results of this paper are not complexity-theoretic hardness results, probabilistic consensus analyses, or cryptographic impossibility proofs. The contribution is a *structural characterization*: we identify a minimal set of resource properties whose presence or absence determines whether adversarial identity cost scales sublinearly or linearly with sustained participation. The novelty lies not in the asymptotic machinery itself, but in the *isolation of sufficient structural conditions* that separate amortizable from throughput-bounded identity regimes—a distinction that existing aggregate-resource models, admission-based defenses, and economic equilibrium analyses do not capture. Prior to this framework, to the best of our knowledge, no unified abstraction classified heterogeneous Sybil-resistance mechanisms (PoW, PoS, device-bound execution, human participation) according to the asymptotic identity-scaling behavior they imply. The framework provides this classification.

Under an amortized coordination model, we establish an asymptotic separation between two broad resource classes:

- Resources satisfying divisibility, transferability, additivity of influence, and temporal reuse admit identity amortization within the considered abstraction, yielding

$$C(s, T) = o(sT).$$

- Resources that are throughput-bounded, non-transferable, and window-local induce the regime

$$C(s, T) = \Omega(sT).$$

Within the present framework, the resulting scaling behavior is primarily governed by the structural properties of the underlying resource class rather than protocol syntax alone. Divisible reusable resources may permit adversarial cost amortization, whereas throughput-bounded participation constraints resist identity reuse across time windows. Importantly, hybrid systems combining both classes inherit the linear lower bound whenever a throughput-bounded component is present (Lemma 5), making this result practically relevant for governance and access-control protocol design.

The framework is intended as a complementary analytical lens alongside protocol-level and economic security analyses.

C. Contributions

- **Axiomatic Resource Framework.** We introduce a framework based on structural resource properties including divisibility, transferability, temporal reuse, additivity of influence, and throughput boundedness.

- **Asymptotic Characterization of Divisible Resources.** We establish that fully divisible and reusable resources admit identity amortization within the considered model, yielding $C(s, T) = o(sT)$.
- **Linear Scaling Bound for Throughput-Limited Resources.** We establish that throughput-bounded, non-transferable participation resources imply $C(s, T) = \Omega(sT)$.
- **Hybrid Resource Composition (Lemma 5).** For systems combining divisible and throughput-bounded resource components, we show that the throughput-bounded component dominates asymptotically under additive decomposition, preserving the linear lower bound. This result is practically relevant for hybrid governance and access-control protocols.
- **Scaling Separation.** We characterize an asymptotic separation between the two resource classes under the assumptions formalized in this work.
- **Implications for Open-System Design.** The framework highlights how identity-level security depends jointly on protocol rules and the structural properties of the underlying security resource.

D. Organization of the Paper

Section II reviews related work. Section III introduces the system model. Section IV formalizes the resource framework. Sections V and VI establish the main asymptotic bounds. Section VII presents the scaling separation theorem. Section VIII discusses concrete resource instantiations and provides lightweight systems anchoring. Sections IX and X examine adversarial implications and limitations. Section XI presents analytical illustrations of the scaling regimes.

II. RELATED WORK

Prior research has addressed adversarial influence and Sybil attacks in open systems from several perspectives. However, these research directions typically study security questions that differ from the one considered in this paper. In particular, existing work largely focuses on identity admission, consensus safety, or economic incentives, while the present work studies how the *structural properties of security resources determine the asymptotic cost of sustaining multiple identities over time*.

Table I summarizes the distinction between major research directions and the focus of this paper.

A. Resource-Based Sybil Defenses

The Sybil attack, formally introduced by Douceur [1], established that identity replication in open systems cannot be prevented without binding influence to scarce resources. Modern permissionless systems implement this principle through resource-weighted mechanisms.

Nakamoto’s proposal [29] introduced Proof-of-Work (PoW), in which block-production probability scales with computational effort. Subsequent formal analyses [5], [6], [19], [20] established security properties of Nakamoto-style consensus, including chain growth, common prefix, and chain quality guarantees.

TABLE I
POSITIONING OF THIS WORK RELATIVE TO PRIOR LITERATURE

Research Line	Primary Security Question	Representative Works
Sybil Attack Literature	Can adversaries create multiple identities, and how can identity creation be restricted or detected?	[1], [14]–[18]
Blockchain Consensus Security	Under what conditions does aggregate adversarial resource ownership violate safety or liveness properties of consensus protocols?	[5], [6], [11], [19]–[21]
Economic Analyses of Consensus	What equilibrium incentives and corruption costs arise in resource-weighted consensus systems?	[22]–[26]
Identity-Binding Mechanisms	How can systems enforce one-person–one-identity mappings through ceremonies or identity verification?	[27], [28]
This Paper	How do structural properties of security resources determine the asymptotic cost of sustaining multiple identities over time?	Structural analysis of identity-scaling cost

Proof-of-Stake (PoS) systems replace computational effort with locked financial capital. Protocols such as Ouroboros [11] and its extensions [30], [31] provide provable security guarantees under partially synchronous network models, while protocols such as Algorand [32], [33] use cryptographic sortition to sample committees proportionally to stake.

Empirical and systems-level analyses further study the security and performance of blockchain consensus mechanisms. Work on mining incentives, pooling behavior, and adversarial strategies [25], [26], [34], [35] demonstrates how adversarial resources are frequently aggregated, pooled, and redistributed across multiple logical identities. Other studies analyze decentralization and deployment realities [7], [9], [36], [37].

Additional work studies network-level adversaries such as eclipse and routing attacks [38]–[40], as well as economic stability of mining rewards and fee mechanisms [22]–[24], [41].

Despite this extensive literature, these analyses consistently reason about *aggregate adversarial resource share*. Security guarantees are expressed in terms of the fraction of computational power or stake controlled by an adversary. However, these models do not explicitly analyze how adversarial cost scales with the number of sustained identities.

B. Graph-Based and Social-Structure Defenses

Another line of work constrains Sybil proliferation using the structure of social networks. Mechanisms such as SybilGuard [14] and SybilLimit [15] exploit the limited number of attack edges between honest and adversarial regions of a social graph to restrict the influence of Sybil identities.

Later work such as SybilInfer [16] applies probabilistic inference over trust graphs to detect clusters of suspicious nodes. These approaches can significantly increase the difficulty of creating large Sybil regions when the social graph exhibits strong connectivity properties.

However, their guarantees depend heavily on the topology and trust assumptions of the underlying network. When such assumptions fail, these mechanisms do not provide explicit

bounds on how adversarial cost scales with the number of sustained identities. More generally, these approaches constrain the *placement* of identities within a social graph, but do not characterize the *cost of sustaining identities* as their number grows.

C. Randomness, Committees, and Human-Centric Mechanisms

Cryptographic randomness primitives such as VRFs [42] and bias-resistant beacons [43] improve scalability and fairness of leader selection. Composable blockchain analyses [5], [19], [44] and asynchronous protocols such as HoneyBadgerBFT [45] extend these guarantees under weaker timing assumptions. However, these mechanisms model adversarial capability in terms of aggregate resource ownership; randomness primitives do not alter the divisibility or transferability of the underlying resource.

Human-centric mechanisms such as Proof-of-Personhood [27] and pseudonym parties [28] bind credentials to real-world participants to restrict identity creation. These approaches address identity admission rather than sustained participation cost; once admitted, influence may still be governed by transferable resources.

Capability architectures [46], congestion control [47], and hardware-rooted execution environments such as Intel SGX [48] impose per-actor throughput limits, but are typically analyzed in terms of access control or network stability rather than adversarial identity-scaling cost.

D. Positioning

Unlike prior work on consensus thresholds, cost-of-corruption, or identity admission, this paper asks whether a fixed aggregate resource can be decomposed across many identities at sublinear marginal cost. We formalize divisibility, temporal reusability, identity transferability, and throughput boundedness as the properties that primarily govern this question, and establish an asymptotic boundary between resource classes. The framework is intended to complement, not replace, existing protocol-level and economic analyses.

TABLE II
CORE NOTATION

Symbol	Meaning
v	Identity (participant instance)
t	Time window index
T	Time horizon (number of windows)
s	Number of sustained identities
$r_v(t)$	Resource allocated to identity v at time t
$W_v(t)$	Influence of identity v at time t
$f(\cdot)$	Resource-to-influence mapping
S_t	Set of adversarial identities at time t
$C(s, T)$	Cost to sustain s identities over T windows
$\Delta(s, T)$	Marginal identity cost

III. SYSTEM MODEL

We consider an open and permissionless system in which identities are inexpensive to create and no trusted admission authority exists. Participation is therefore identity-unbounded, and adversaries may generate arbitrarily many identities at negligible syntactic cost.

Security mechanisms regulate influence by binding participation to the possession or expenditure of a resource R . Our objective is to characterize how structural properties of R determine the scaling behavior of adversarial identity replication.

A. Notation

Table II summarizes the core notation used throughout the paper.

B. Open Identity Model

Let $\mathcal{U}$ denote the universe of potential participants. At each time window t , the system contains a set of active identities V_t .

An adversary $\mathcal{A}$ is identity-unbounded: it may create, retire, and coordinate an arbitrary number of identities at negligible syntactic cost, consistent with the Sybil model [1].

Let $S_t \subseteq V_t$ denote the subset of identities controlled by $\mathcal{A}$ during window t , with $|S_t| = s_t$. We impose no a priori bound on s_t .

An identity is an abstract participant instance capable of receiving resource allocation and producing influence.

C. Time Model

Time is divided into discrete windows

$$t = 1, 2, \dots$$

A window represents the minimal interval over which resource allocation and influence may be refreshed (e.g., block intervals or governance epochs).

We evaluate security over a finite horizon of T windows. An identity is said to be *sustained* if it remains active throughout all windows $t = 1, \dots, T$.

D. Resource and Influence

Let R denote the security-weighting resource. For each identity $v \in V_t$ and window t ,

$$r_v(t) \in \mathbb{R}_{\geq 0}$$

denotes the allocated resource. Influence is determined by a mapping

$$W_v(t) = f(r_v(t)).$$

We do not impose structural assumptions on f at this stage; these will be derived from properties of the resource R .

Definition 1 (Activation Threshold). *There exists a constant $r_{\min} > 0$ such that an identity is active in window t only if*

$$r_v(t) \geq r_{\min}.$$

This captures the requirement that participation requires a strictly positive resource commitment.

Remark 1. *The activation threshold $r_{\min}$ establishes a per-identity floor that any active identity must meet, regardless of resource structure. This baseline cost is unavoidable and grows linearly with the number of sustained identities. The structural separation established in Sections V–VII does not concern this baseline. Rather, it asks whether a resource imposes an additional per-identity, per-window cost beyond $r_{\min}$. Specifically, Theorem 1 shows that a fully parallelizable resource may not impose such additional cost within this model: an adversary need only acquire an aggregate allocation $r \geq s r_{\min}$ and redistribute it across s identities, incurring no further T -scaling overhead. Theorem 2 shows that a throughput-bounded resource does impose this additional structure, making $r_{\min} \cdot T$ the unavoidable marginal cost per new identity.*

The total adversarial influence at time t is

$$W_{\mathcal{A}}(t) = \sum_{v \in S_t} W_v(t).$$

E. Adversarial Model

The adversary $\mathcal{A}$ is computationally unbounded but constrained by the structural properties of the resource R .

Specifically, $\mathcal{A}$ may:

- Instantiate arbitrarily many identities.
- Allocate and redistribute resource across identities when permitted by the resource structure.
- Reuse or reallocate resource across time windows when temporal reuse is allowed.
- Coordinate identities without communication constraints.

This model subsumes aggregate-resource adversaries used in blockchain analyses [5], [11], while remaining protocol-independent.

F. Model Scope and Coordination Assumptions

Our adversarial model assumes that identity coordination does not itself impose throughput constraints comparable to those enforced by the underlying security resource.

In particular, we distinguish between: (i) resource-induced constraints, which arise from the structural properties of the security-weighting resource, and (ii) coordination overhead, which captures auxiliary costs associated with managing multiple identities.

Throughout the paper, coordination costs are modeled separately through the function $h(s, T)$, allowing the structural properties of the resource to be analyzed independently of implementation-specific orchestration mechanisms.

The results established in Sections V–VII therefore isolate the scaling implications induced by resource structure rather than by assumptions about communication complexity, software architecture, or operational deployment details.

G. Adversarial Cost

We define the adversarial cost function $C(s, T)$ as the minimum total resource expenditure required to sustain at least s active identities over T consecutive windows.

To make the structure of costs precise, we decompose $C(s, T)$ into three components:

$$C(s, T) = C_{\text{stock}}(s) + C_{\text{flow}}(s, T) + h(s, T),$$

where:

- $C_{\text{stock}}(s)$ is the one-time acquisition cost of resources that persist across all T windows without renewal (e.g., staked capital, hardware);
- $C_{\text{flow}}(s, T)$ is the cumulative per-window expenditure for resources that must be renewed each window (e.g., energy, device attestations, human participation);
- $h(s, T)$ is the coordination overhead of managing s identities over T windows.

Formally,

$$C(s, T) = \inf \left\{ \sum_{t=1}^T \mathcal{R}_{\mathcal{A}}(t) \mid |S_t| \geq s \ \forall t \in \{1, \dots, T\} \right\}.$$

The structural separation of Sections V–VII concerns the scaling of $C_{\text{flow}}(s, T)$: parallelizable resources admit $C_{\text{flow}}(s, T) = 0$ (stock suffices for all windows), whereas throughput-bounded resources enforce $C_{\text{flow}}(s, T) = \Omega(sT)$. The coordination term $h(s, T)$ is governed by Assumption 1.

H. Scaling Regimes

We distinguish two asymptotic regimes:

- **Sublinear Scaling:**

$$C(s, T) = o(sT)$$

- **Linear Scaling:**

$$C(s, T) = \Omega(sT)$$

Linear scaling implies that each additional sustained identity incurs proportional per-window cost. Sublinear scaling implies identity amortization, where marginal identity cost vanishes asymptotically.

I. Objective

Our goal is to characterize how structural properties of the resource R determine which scaling regime arises.

In particular, we show that adversarial identity scaling is shaped not by protocol design alone, but by intrinsic structural constraints of the underlying resource.

IV. AXIOMATIC THEORY OF SECURITY RESOURCES

We develop an axiomatic framework for security-weighting resources that isolates the structural properties responsible for adversarial identity scaling.

Rather than analyzing specific protocols, we model the underlying resource R used to assign influence to identities. Our goal is to characterize how intrinsic structural properties of R determine whether identity amplification is possible.

Throughout the analysis, let $r_v(t) \in \mathbb{R}_{\geq 0}$ denote the resource allocation assigned to identity v during window t , and let $W_v(t) = f(r_v(t))$ denote the resulting influence.

A. Resource as a Structural Object

We model a security-weighting resource as a quantity that

- 1) can be allocated across identities,
- 2) evolves across time windows, and
- 3) determines influence through an allocation rule f .

Our analysis is agnostic to the semantic nature of the resource (e.g., computation, capital, or attention). Instead, it depends only on structural properties governing division, aggregation, reuse, and transfer across identities and time.

B. Structural Properties

Definition 2 (Divisibility). *A resource R is divisible if for any allocation $r \geq 0$ and integer $k \geq 1$ there exist allocations $r_1, \dots, r_k \geq 0$ such that*

$$\sum_{i=1}^k r_i = r,$$

and the adversary may distribute r_i across k identities without additional structural cost.

Definition 3 (Additivity of Influence). *A resource induces additive influence if*

$$\sum_{v \in S_t} f(r_v(t)) = f\left(\sum_{v \in S_t} r_v(t)\right).$$

Definition 4 (Temporal Reusability). *A resource is temporally reusable if allocation in window t may be reused in window $t + 1$ without renewed identity-bound expenditure.*

Definition 5 (Identity Transferability). *A resource is identity-transferable if an allocation $r_v(t)$ may be reassigned to another identity v' without proportional cost.*

Definition 6 (Throughput-Bounded Resource). *A resource R is throughput-bounded if participation is mediated through independent execution channels, each bound to a specific identity such that resource cannot be reassigned across identities without proportional cost.*

Formally, each identity v is associated with a dedicated channel C_v satisfying all three of the following conditions:

- 1) **Per-channel rate limit:** there exists a constant $\tau > 0$ such that

$$r_v(t) \leq \tau$$

for all v and t ;

- 2) **Non-transferability:** the allocation $r_v(t)$ assigned to identity v cannot be simultaneously credited to any other identity $v' \neq v$;
- 3) **Window-locality:** allocation in window t does not carry over to window $t + 1$; sustained participation therefore requires renewed expenditure in every window.

Parameter relationship. The throughput bound τ and the activation threshold $r_{\min}$ (Definition 1) are independent parameters governing distinct structural constraints. The activation threshold specifies the minimum resource required for an identity to be active in a given window, while τ bounds the maximum resource any single identity can contribute per window. For the system to admit active identities, these parameters must satisfy $0 < r_{\min} \leq \tau$; otherwise no identity can meet the activation threshold within its channel capacity. The linear lower bound of Theorem 2 holds for any such configuration and is tightest when $r_{\min} = \tau$.

Remark 2 (Saturated channels: $r_{\min} = \tau$). The boundary case $r_{\min} = \tau$ corresponds to fully saturated participation channels: each active identity consumes the entire per-window throughput allocation of its dedicated channel. In this setting no capacity slack exists and any additional identity strictly requires a new independent channel. The lower bound $C(s, T) \geq sTr_{\min}$ becomes tight, and the linear scaling regime holds with constant 1.

Throughput boundedness imposes a per-identity rate constraint that is structurally non-transferable and window-local. Scaling the number of identities therefore requires provisioning additional independent channels.

C. Structural Implications

The above properties jointly determine adversarial scaling behavior.

Proposition 1 (Identity Partition Invariance Under Parallelizable Resources). *If a resource R satisfies divisibility, additivity, temporal reusability, and identity transferability, then adversarial influence depends only on aggregate resource ownership and is independent of identity partitioning.*

Proof. Let R satisfy all four properties and let A control aggregate allocation $r = \sum_{v \in S_t} r_v(t)$. By divisibility, r may be partitioned across any set of identities; by additivity, $\sum_v f(r_v) = f(\sum_v r_v) = f(r)$, so total influence $W_A(t) = f(r)$ is invariant to partitioning. By identity transferability, any reallocation of r across identities leaves $W_A(t)$ unchanged. By temporal reusability, r acquired in window t persists across all subsequent windows without renewed expenditure. Hence $W_A(t) = f(r)$ depends only on aggregate ownership, independent of identity configuration. $\square$

D. Resource Classes

Definition 7 (Parallelizable Resource). *A resource R is parallelizable if it satisfies divisibility, additivity of influence, temporal reusability, and identity transferability.*

Definition 8 (Throughput-Bounded Non-Parallelizable Resource). *A throughput-bounded non-parallelizable resource is any resource R satisfying Definition 6.*

E. Resource Taxonomy

Table III summarizes representative resource classes and the structural properties they satisfy. Proof-of-Work is treated as two distinct components: the reusable *capacity* component (mining hardware) and the non-reusable *operational* component (energy expenditure). The structural results of this paper apply to each component separately; the full economic cost of PoW mining involves both and must be analyzed accordingly.

Parallelizable resources admit subdivision and reuse of their stock component across identities and time. PoW is hybrid: its capacity component (hardware) is reusable stock (Theorem 1 applies); its operational component (energy) is window-local flow and must be renewed each window. Throughput-bounded resources require one independent channel per identity, yielding linear cost scaling.

F. Coordination Cost Assumption

Assumption 1 (Amortized Coordination). *The adversarial cost of coordinating s identities over T windows satisfies*

$$h(s, T) = o(sT).$$

Scope. This is an environmental and deployment assumption, not a structural property of the security-weighting resource R . It models the operational overhead of managing multiple identities—such as software orchestration, account administration, or communication—rather than the resource-induced participation cost captured by $C_{\text{flow}}(s, T)$. The two components are independent: $h(s, T)$ can be reduced through automation or shared infrastructure, whereas $C_{\text{flow}}(s, T)$ is determined solely by the structural properties of R .

Justification. A representative sufficient instance is $h(s, T) = O(s + T)$, which models shared infrastructure—such as cloud orchestration platforms, automated scripting frameworks, or centralized delegation services—whose cost scales with the number of managed identities and the duration of deployment, but not with the product sT . This condition is consistent with empirically observed pooling behavior in deployed systems [7], [12], [13]. Examples include mining pools, validator delegation, and cloud-managed multi-account orchestration.

Role in the analysis. Theorems 1 and 2 isolate resource-induced scaling independently of $h(s, T)$: Theorem 1 holds even when $h(s, T) = 0$, and Theorem 2 holds regardless of how small $h(s, T)$ is.

Remark 3 (Boundary case: $h(s, T) = \Theta(sT)$). *If coordination overhead itself scales linearly with total sustained participation, i.e., $h(s, T) = \Theta(sT)$, then coordination becomes the dominant cost term and the total adversarial cost satisfies*

TABLE III
STRUCTURAL TAXONOMY OF SECURITY RESOURCES. PROOF-OF-WORK IS DECOMPOSED INTO ITS CAPACITY COMPONENT (MINING HARDWARE: PARALLELIZABLE STOCK) AND ITS OPERATIONAL COMPONENT (ENERGY: NON-REUSABLE FLOW). THEOREM 1 APPLIES TO THE IDENTITY-PARTITIONING BEHAVIOR OF THE CAPACITY COMPONENT ONLY. THE FULL ECONOMIC COST OF POW INCLUDES BOTH COMPONENTS.

Resource Class	Div.	Add.	Reuse	Transfer	Cost Type	Scaling
PoW: Capacity component (hardware)	✓	✓	✓	✓	Stock	Sublinear
PoW: Operational component (energy)	✓	✓	×	✓	Flow	Sublinear in s , Linear in $T^\dagger$
Capital (PoS)	✓	✓	✓	✓	Stock	Sublinear
Social-Graph Trust	Partial	Partial	Partial	Partial	Mixed	Model-Dependent
Human Real-Time Channel	×	×	×	×	Flow	Linear
Device-Bound Execution	×	×	×	×	Flow	Linear

[†] Energy expenditure is non-reusable across windows (Temporal Reuse = $\times$) but remains transferable across identities and divisible. It therefore produces a flow cost that scales linearly in T (each window requires fresh expenditure) but sublinearly in s (energy can be redistributed across identities). This component falls outside the purely parallelizable class of Theorem 1, which applies only to the capacity (stock) component of PoW.

$C(s, T) = \Theta(sT)$ even for a divisible, reusable resource. In this regime, orchestration overhead effectively reintroduces throughput-bounding structure at the coordination layer. The framework therefore applies to settings in which coordination infrastructure allows sublinear overhead; when it does not, the system exhibits linear scaling by a different mechanism.

G. Identity-Amortization Property

Definition 9 (Identity-Amortizable Mechanism). A mechanism is identity-amortizable if there exists a function $g(s, T)$ such that

$$g(s, T) = o(sT) \quad \text{and} \quad C(s, T) \leq g(s, T)$$

for sufficiently large s .

Identity amortization implies

$$\lim_{s \rightarrow \infty} \frac{C(s, T)}{sT} = 0.$$

H. Structural Implication

The remainder of the paper establishes that divisible, reusable resources admit $C(s, T) = o(sT)$, while rate-limited resources enforce $C(s, T) = \Omega(sT)$. Adversarial identity scaling thus reduces to a classification problem over the resource R .

V. STRUCTURAL LIMITS OF PARALLELIZABLE RESOURCES

We establish that divisible, reusable resources admit identity amortization under the coordination model of Assumption 1.

A. Marginal Identity Cost

To study identity-level scaling, we define the marginal sustained identity cost

$$\Delta(s, T) = C(s, T) - C(s-1, T).$$

A mechanism exhibits *vanishing marginal cost* if

$$\lim_{s \rightarrow \infty} \Delta(s, T) = 0.$$

Vanishing marginal cost implies identity amortization in a strong sense: each additional sustained identity becomes asymptotically free.

B. Adversarial Amortization Strategy

Let R be a parallelizable resource. By Definition 7, R satisfies divisibility, additivity of influence, temporal reusability, and identity transferability.

We construct an adversarial strategy that sustains s identities over T windows while keeping total resource expenditure independent of s , up to coordination overhead.

C. Identity Partition Invariance

Lemma 1. Let R be divisible and additive. For any aggregate allocation $r \geq 0$ and any integer $s \geq 1$, the adversary can distribute r across s identities such that

$$\sum_{v=1}^s f(r_v) = f(r) \quad \text{and} \quad \sum_{v=1}^s r_v = r.$$

Proof. By divisibility, r may be partitioned into allocations $r_1, \dots, r_s$ such that $\sum_{v=1}^s r_v = r$. By additivity,

$$\sum_{v=1}^s f(r_v) = f\left(\sum_{v=1}^s r_v\right) = f(r).$$

Thus influence is invariant to identity partitioning. $\square$

D. Cross-Window Reuse

Lemma 2. If R is temporally reusable, a resource allocation r obtained in window t may be reused in all subsequent windows $t+1, \dots, T$ without additional expenditure.

Proof. This follows directly from the definition of temporal reusability. $\square$

E. Coordination Overhead

The preceding arguments show that aggregate resource allocation can be reused across identities and time. However, maintaining multiple identities may incur additional coordination overhead. Let $h(s, T)$ denote the adversarial cost of coordinating s identities over T windows. The scaling behavior of $h(s, T)$ is governed by Assumption 1, stated in Section IV. The structural results are robust to any weaker condition $h(s, T) = o(sT)$, as noted therein.

F. Marginal Cost Collapse

Lemma 3. *Under the amortized coordination model, as $s, T \rightarrow \infty$ jointly,*

$$\frac{C(s, T)}{sT} \rightarrow 0,$$

and the marginal identity cost satisfies

$$\Delta(s, T) = O(1) = o(T).$$

Proof. By the activation threshold (Definition 1), sustaining s identities requires aggregate allocation at least $r \geq s r_{\min}$. For a parallelizable resource, the adversary acquires this aggregate *once*: by temporal reusability (Lemma 2), the allocation $r = s r_{\min}$ suffices for all T windows without renewed expenditure. By Lemma 1, r is then partitioned across s identities without changing total influence.

Thus the total adversarial cost satisfies

$$C(s, T) \leq s r_{\min} + h(s, T).$$

Under Assumption 1, $h(s, T) = o(sT)$. Therefore

$$\frac{C(s, T)}{sT} \leq \frac{r_{\min}}{T} + \frac{h(s, T)}{sT} \rightarrow 0 \quad \text{as } s, T \rightarrow \infty,$$

since the first term vanishes as $T \rightarrow \infty$ and the second by assumption. Hence $C(s, T) = o(sT)$.

For the marginal cost,

$$\Delta(s, T) \leq r_{\min} + [h(s, T) - h(s-1, T)] = O(1),$$

which is $o(T)$, confirming sublinear marginal cost. $\square$

G. Main Structural Limit Theorem

Theorem 1 (Structural Limit of Parallelizable Resources). *Let R be a parallelizable resource. Under the amortized coordination model of Assumption 1, any mechanism grounded in R is identity-amortizable within the present abstraction:*

$$C(s, T) = o(sT).$$

In particular, the marginal cost per additional identity satisfies $\Delta(s, T) = o(T)$: it does not scale with the time horizon.

Proof. By Lemma 3, under Assumption 1, as $s, T \rightarrow \infty$ jointly,

$$C(s, T) = o(sT).$$

Therefore any mechanism grounded in R is identity-amortizable by Definition 9. The marginal bound $\Delta(s, T) = o(T)$ follows directly from Lemma 3. $\square$

Remark 4. *Theorem 1 concerns the absence of T -scaling renewal cost: aggregate resource must still grow with s (each identity requires at least $r_{\min}$), but temporal reusability eliminates per-window overhead once the stock is acquired. Introducing $C(s, T) = \Omega(sT)$ under this model requires a non-parallelizable structural component.*

VI. LINEAR LOWER BOUND FOR THROUGHPUT-BOUNDED RESOURCES

We now establish the complementary structural result: throughput-bounded, non-transferable resources enforce linear sustained identity cost.

In contrast to parallelizable resources, where identities can be amortized across time and accounts, throughput-bounded resources require independent execution capacity per identity. This structural constraint prevents identity amplification and forces adversarial cost to scale linearly with both the number of identities and the time horizon.

The result provides a formal foundation for mechanisms that achieve strong Sybil resistance through resource-induced constraints.

A. Structural Constraints

Let R be a throughput-bounded, non-transferable resource with window-local allocation.

Throughput boundedness implies the existence of a constant $\tau > 0$ such that for every identity v and window t

$$r_v(t) \leq \tau.$$

Thus a single identity cannot contribute more than τ units of resource per window.

We further assume that identity transferability fails: resource allocated to identity v cannot be reassigned to another identity without proportional cost.

Finally, resource allocation is window-local, meaning that allocation in window t does not automatically carry over to window $t+1$. Sustained participation therefore requires renewed resource expenditure in every window.

Together, these constraints imply that each identity must be supported by an independent execution channel whose capacity cannot be shared or reused across identities or time.

B. Marginal Identity Cost

Recall the marginal sustained identity cost

$$\Delta(s, T) = C(s, T) - C(s-1, T).$$

Lemma 4 (Positive Marginal Cost Under Throughput Constraints). *Let R be throughput-bounded, non-transferable, and window-local. Then there exists a constant $c > 0$ such that for all sufficiently large s*

$$\Delta(s, T) \geq cT.$$

Proof. To sustain s active identities, each identity must receive at least $r_{\min} > 0$ resource units per window by the activation threshold.

Thus sustaining s identities over T windows requires total allocation at least

$$sTr_{\min}.$$

Similarly sustaining $s-1$ identities requires

$$(s-1)Tr_{\min}.$$

Therefore

$$\Delta(s, T) = C(s, T) - C(s-1, T) \geq Tr_{\min}.$$

Let $c = r_{\min}$. Then

$$\Delta(s, T) \geq cT.$$

□

C. Main Lower Bound Theorem

Theorem 2 (Linear Lower Bound). *Let R be throughput-bounded, non-transferable, and window-local. Then there exists a constant $c > 0$ such that*

$$C(s, T) \geq csT.$$

In particular

$$C(s, T) = \Omega(sT)$$

and the marginal sustained identity cost satisfies

$$\liminf_{s \rightarrow \infty} \Delta(s, T) \geq cT.$$

Proof. Fix s and T ; suppose the adversary sustains s active identities across windows $t = 1, \dots, T$. By the activation threshold (Definition 1), each identity v requires $r_v(t) \geq r_{\min}$ in every active window. Non-transferability (Definition 6, condition 2) prevents $r_v(t)$ from being credited to any other identity, so $\sum_{v=1}^s r_v(t) \geq sr_{\min}$ for each t . Window-locality (condition 3) prevents carry-over, forcing fresh expenditure each window. Summing over T windows:

$$C(s, T) \geq \sum_{t=1}^T sr_{\min} = sTr_{\min}.$$

Setting $c = r_{\min}$ gives $C(s, T) = \Omega(sT)$. The marginal lower bound $\liminf_{s \rightarrow \infty} \Delta(s, T) \geq cT$ follows from Lemma 4. □

VII. EXTREMAL STRUCTURAL SEPARATION

We now establish the extremal scaling separation. The result suggests that adversarial scaling behavior is primarily governed by the structural properties of the underlying resource.

A. Adversarial Scaling Regimes

Definition 10 (Scaling Regime). *A mechanism induces*

- Sublinear scaling if $C(s, T) = o(sT)$,
- Linear scaling if $C(s, T) = \Omega(sT)$.

The scaling regime characterizes how the adversarial cost of sustaining identities grows with the number of identities s and the time horizon T . It therefore determines whether identity amplification is feasible or structurally constrained.

B. Structural Separation Theorem

Theorem 3 (Extremal Structural Separation). *Let R be a security-weighting resource.*

- 1) *If R is parallelizable, the induced mechanism exhibits sublinear scaling.*
- 2) *If R is throughput-bounded, non-transferable, and window-local, the induced mechanism exhibits linear scaling.*

Consequently, parallelizable resources and throughput-bounded non-parallelizable resources induce asymptotically disjoint scaling regimes.

Proof. Part (1) follows directly from the Structural Limit Theorem established in the previous section: parallelizable resources admit identity amortization, implying

$$C(s, T) = o(sT).$$

Part (2) follows from the Linear Lower Bound Theorem: throughput-bounded, non-transferable, window-local resources enforce

$$C(s, T) = \Omega(sT).$$

Since $o(sT)$ and $\Omega(sT)$ represent asymptotically distinct growth regimes, no mechanism can simultaneously exhibit both.

Therefore the two structural resource classes induce disjoint asymptotic scaling regimes. □

Thus, parallelizable resources enable identity amplification, whereas throughput-bounded resources resist, within the considered model, it.

Corollary 1. *Within the structural model of this paper, linear sustained identity cost—and therefore resistance to identity amplification—cannot be enforced within the class of fully parallelizable resources under amortized coordination.*

C. Resource Substitution Principle

The separation result above yields a general design constraint for security mechanisms in open distributed systems.

Theorem 4 (Resource Substitution Principle). *Suppose a security mechanism aims to enforce linear sustained identity cost*

$$C(s, T) = \Omega(sT).$$

Then the underlying security-weighting resource cannot satisfy all structural properties of parallelizability.

In particular, at least one of the following properties must be violated:

- Divisibility,
- Additivity of influence,
- Temporal reusability,
- Identity transferability.

Proof. Assume for contradiction that the underlying resource satisfies all four properties above. By Definition 7, the resource is parallelizable.

From the Structural Limit Theorem, mechanisms grounded in parallelizable resources satisfy, within the present model,

$$C(s, T) = o(sT).$$

This contradicts the assumption that the mechanism enforces

$$C(s, T) = \Omega(sT).$$

Therefore any mechanism achieving linear sustained identity cost must violate at least one structural property defining parallelizability. $\square$

This principle can be interpreted as a design consideration: preventing identity amplification requires introducing structural limitations that break parallelizability.

The theorem highlights a fundamental design rule: linear identity-level Sybil resistance cannot be achieved through protocol design alone under the considered assumptions. Instead it requires selecting a resource class that structurally restricts parallelization, transferability, or temporal reuse.

D. Intermediate Structural Regimes

The separation established in Theorem 3 characterizes two extreme structural classes. However, real-world mechanisms may satisfy structural properties only partially, yielding scaling behavior that interpolates between the two asymptotic regimes established in this work.

Partial Transferability: Suppose identity transferability holds only up to a bounded fraction $\alpha \in [0, 1]$ of the aggregate allocation. That is, at most $\alpha \cdot r$ units of resource may be reassigned across identities, while the remaining $(1-\alpha) \cdot r$ units are identity-bound. In this case, the adversary can amortize only a fraction of the aggregate cost, yielding a mixed scaling regime. Specifically, one would expect:

$$C(s, T) = \Omega((1 - \alpha) \cdot sT),$$

with the bound tightening as $\alpha \rightarrow 0$ (approaching full throughput-boundedness) and weakening as $\alpha \rightarrow 1$ (approaching full parallelizability).

Bounded Temporal Reuse: Suppose resource allocation persists across at most k consecutive windows rather than indefinitely. An adversary must then renew allocation every k windows, incurring cost proportional to $\lceil T/k \rceil$ renewal events. The resulting adversarial cost satisfies:

$$C(s, T) = \Omega\left(s \cdot \frac{T}{k}\right),$$

interpolating between linear scaling at $k = 1$ (window-local, full renewal) and sublinear scaling as $k \rightarrow T$ (full temporal reuse).

General Intermediate Regime: More generally, partial violations of parallelizability may yield cost functions satisfying

$$\omega(1) \leq C(s, T) \leq o(sT),$$

or other mixed asymptotic forms depending on which structural properties are partially relaxed and to what degree.

Characterizing these intermediate scaling regimes in full generality remains an open theoretical problem. A natural extension would reformulate the axiomatic framework using continuous structural parameters $(\alpha_{\text{div}}, \alpha_{\text{add}}, \alpha_{\text{reuse}}, \alpha_{\text{transfer}}) \in [0, 1]^4$ rather than binary property classes, and derive adversarial cost bounds as functions of these parameters.

The present work therefore establishes the extremal boundary conditions under which identity amortization is either unavoidable or asymptotically eliminated, leaving the interior of this parameter space as a direction for future work.

VIII. INSTANTIATIONS OF THROUGHPUT-BOUNDED RESOURCES

This section characterizes mechanisms that instantiate the throughput-bounded resource class of Sections VI–VII and illustrates how the resulting structural constraints influence adversarial identity scaling. Figure 1 provides a conceptual overview of the relationship between resource structure and scaling behavior.

A. Per-Actor Throughput Channels

Let $\mathcal{U}$ denote a set of independent actors. Each actor $u \in \mathcal{U}$ is associated with a real-time execution channel $\mathcal{C}_u$ through which resource output is generated.

Definition 11 (Per-Actor Throughput Channel). *A channel $\mathcal{C}_u$ is throughput-bounded if there exists a constant $\tau > 0$ such that for every window t*

$$\text{rate}_t(\mathcal{C}_u) \leq \tau.$$

The quantity $\text{rate}_t(\mathcal{C}_u)$ represents the maximum resource output attributable to actor u during window t . Resource allocation for identity v is derived from the channel output:

$$r_v(t) \leq \text{rate}_t(\mathcal{C}_u).$$

This abstraction naturally induces throughput boundedness, identity non-transferability, and window-local temporal scope. Critically, it enforces that each identity must be backed by an independent execution channel, preventing cross-identity resource reuse.

B. Formal Classification

Theorem 5 (Per-Actor Channel Classification). *Let R be a resource derived from per-actor throughput channels. Then R satisfies (1) throughput boundedness, (2) failure of identity transferability, and (3) window-local temporal scope. Consequently R belongs to the throughput-bounded non-parallelizable class.*

Proof. Throughput boundedness follows from the existence of the per-channel rate limit τ . Identity transferability fails

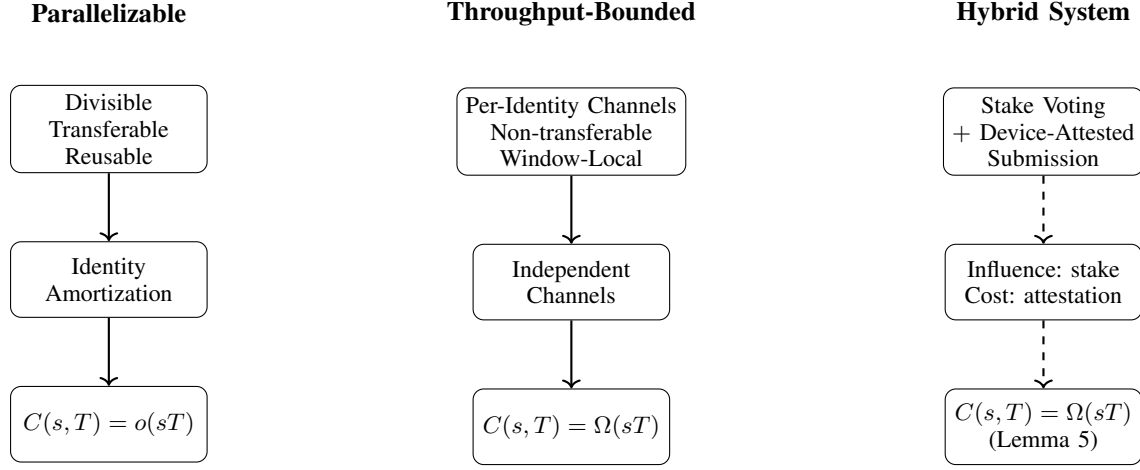


Fig. 1. Structural classification of adversarial identity cost. Parallelizable resources (left) enable identity amortization and sublinear cost. Throughput-bounded resources (center) enforce independent per-identity channels and linear cost. A hybrid governance system (right) combines stake-weighted aggregate influence with device-attested proposal submission; by Lemma 5, under additive decomposition assumptions, the throughput-bounded component dominates and linear adversarial cost is preserved.

because resource generation is bound to the actor controlling channel C_u . Window locality holds since each window requires fresh channel output. Thus the resource satisfies the defining properties of throughput-bounded non-parallelizable resources. $\square$

C. Examples of Throughput-Bounded Channels

Several mechanisms naturally instantiate per-actor throughput channels.

a) *Human Real-Time Participation*: Human cognition and motor interaction impose intrinsic per-window throughput limits that cannot be increased by adding computational infrastructure. Adversarial scaling therefore requires recruiting additional independent actors.

b) *Device-Bound Execution*: Trusted execution environments and device attestation systems bind activity to specific hardware [48]. Each device produces limited verifiable output per window; a single device cannot sustain multiple identities simultaneously.

c) *Rate-Limited Network Participation*: Account-level rate limits qualify as throughput-bounded only when structurally bound to non-transferable execution channels. Limits attached to syntactic identifiers alone are bypassable through identity replication. Effective enforcement requires binding rate constraints to independent, non-aggregable participation channels such as device-attested proposal submission or hardware-rooted governance actions.

D. Hybrid Resource Models

Real-world systems rarely rely on a single security resource in isolation. Instead, deployed protocols often combine parallelizable and throughput-bounded resources. For example, a governance protocol may allocate voting power proportionally to financial stake while limiting proposal submission through per-identity rate constraints.

Lemma 5 (Hybrid Scaling Behavior). *Let $R = (R_p, R_t)$ where R_p is parallelizable and R_t is throughput-bounded and non-transferable. If adversarial cost decomposes as*

$$C(s, T) = C_p(s, T) + C_t(s, T),$$

then

$$C_p(s, T) = o(sT), \quad C_t(s, T) = \Omega(sT).$$

Consequently

$$C(s, T) = \Omega(sT).$$

Proof. Under additive decomposition, $C(s, T) = C_p(s, T) + C_t(s, T)$. By Theorem 1, $C_p(s, T) = o(sT)$. By Theorem 2, $C_t(s, T) \geq c \cdot sT$ for some $c > 0$. For sufficiently large s , the throughput-bounded term dominates: $C(s, T) \geq C_t(s, T) \geq c \cdot sT$, giving $C(s, T) = \Omega(sT)$. $\square$

The additive decomposition is a structural modeling assumption: the two components are assumed to contribute independently. Non-additive compositions remain outside the present scope.

E. Analytical Illustration of the Scaling Laws

Figure 2 plots the closed-form cost expressions for $s \in [1, 10,000]$ and $T = 100$. The parallelizable curve is flat (identity amortization); the throughput-bounded curve grows linearly with s , consistent with $C(s, T) = \Omega(sT)$.

F. Illustrative Systems Anchoring

The structural distinctions formalized in this paper correspond to observable participation patterns in deployed systems. The following illustrate the connection at order-of-magnitude level and are not empirical measurements or model validations.

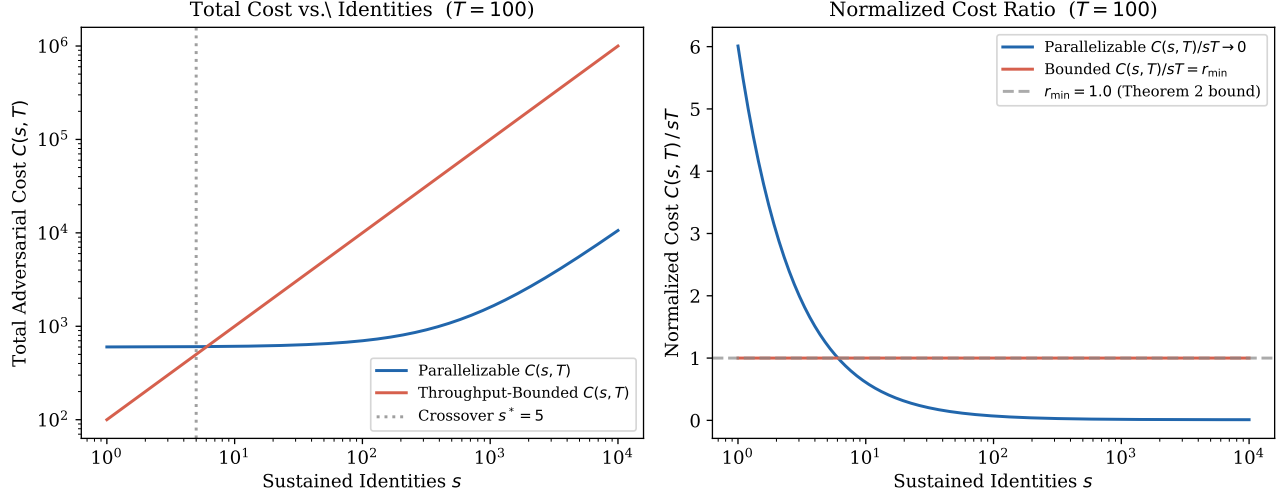


Fig. 2. Adversarial cost scaling derived from Theorems 1–2. *Left*: total adversarial cost $C(s, T)$ vs. sustained identities s ; parallelizable resources exhibit cost growing only with s (stock acquisition) whereas throughput-bounded resources enforce linear growth in both s and T . The crossover $s^* = r_{\text{fixed}}/(r_{\min}T)$ marks where throughput-bounded cost exceeds the parallelizable baseline. *Right*: normalized total cost $C(s, T)/sT$; the parallelizable ratio declines as $s, T \rightarrow \infty$ jointly (Theorem 1), while the throughput-bounded ratio remains bounded away from zero at $r_{\min}$ (Theorem 2). Note that $C(s, T)/sT$ measures amortized cost per identity per window and should not be confused with marginal identity cost $\Delta(s, T) = C(s, T) - C(s - 1, T)$, which for throughput-bounded resources satisfies $\Delta(s, T) \geq r_{\min}T$.

a) *Mining pool concentration (PoW)*.: Empirical studies of Bitcoin and Ethereum (pre-Merge) consistently document that a small number of mining pools account for the majority of block production. Gencer et al. [9] report that the top-4 pools frequently control over 50% of hash power, and similar patterns appear across other PoW networks. This concentration is structurally consistent with the divisibility and transferability of mining capacity: operators aggregate hardware across nominally distinct identities without proportional renewal cost per identity. This is precisely the identity-amortization behavior formalized in Theorem 1 for divisible resources.

b) *Validator delegation in PoS*.: In Ethereum’s proof-of-stake protocol, the majority of staked ETH is concentrated among a small number of liquid staking providers (e.g., Lido, Coinbase, Rocket Pool). As of 2023, Lido Finance alone controlled approximately 30% of staked ETH [49], with a handful of providers jointly exceeding 50%. Delegation allows a fixed capital base to be distributed across many validators, again instantiating the amortization pattern of Theorem 1 for transferable, reusable resources.

c) *Governance proposal rates*.: In on-chain governance systems (e.g., Compound, Uniswap, MakerDAO), proposal submission typically requires meeting a threshold of delegated voting power. An actor who splits their stake across s accounts and submits proposals from each can linearly multiply proposal bandwidth without proportional capital increase — provided no non-transferable per-identity throughput constraint is enforced. This illustrates the vulnerability described in the stake-governed participation case study (Section IX-C).

d) *Device-attested participation*.: Hardware security keys and trusted execution environments (e.g., Intel SGX-based attestation) enforce per-device participation limits. A single device cannot simultaneously generate valid attestations for multiple concurrent identities. In systems where gover-

nance actions require valid hardware attestation per submission, this imposes a structural per-identity throughput limit consistent with Definition 6, approximating the throughput-bounded regime of Theorem 2.

These patterns are cited at the order-of-magnitude level to illustrate that the structural conditions modeled in this paper correspond to observable participation behavior. No quantitative claim about the applicability of the theoretical bounds to specific deployed systems is made.

G. Practical Enforcement of Throughput Bounds

The theoretical results developed in this paper assume the existence of non-transferable per-identity throughput bounds. In practice such constraints are approximated through mechanisms that bind resource generation to independent actors, devices, or execution environments.

Examples include rate-limited protocol participation, device-attested computation, and human-interaction tasks. While no practical mechanism perfectly enforces strict non-transferability, systems approximating these structural properties may approach the linear scaling conditions of Theorem 2.

IX. ADVERSARIAL IMPLICATIONS AND ATTACK SURFACES

The structural separation of Sections V–VII has direct implications for adversarial participation. When a resource is divisible and reusable, capacity may be redistributed across identities without proportional cost growth, shifting the key security question from aggregate ownership to identity-level participation scaling.

A. Canonical Amortization Strategy

Let R be a parallelizable resource. An adversary acquires aggregate allocation $r = s r_{\min}$, partitions it across s identities, and reuses the allocation across windows. By Theorem 1,

$$C(s, T) = o(sT),$$

so the marginal cost of sustaining additional identities does not grow proportionally with the time horizon. In contrast, throughput-bounded resources enforce $\Delta(s, T) = \Omega(T)$ by Theorem 2.

B. Instantiation in Blockchain Systems

a) *Proof-of-Work.*: Proof-of-Work combines a reusable stock component (mining hardware) with a non-reusable flow component (energy expenditure). Mining capacity is divisible and transferable, allowing aggregate hash power to be redistributed across identities through mechanisms such as mining pools. Operational energy expenditure, however, remains window-local and must be renewed continuously.

b) *Proof-of-Stake.*: Financial stake satisfies divisibility, transferability, and temporal reuse. Delegation mechanisms therefore allow aggregate capital to be redistributed across multiple validators without proportional identity-level renewal cost. Under the present model, the stake component admits identity amortization.

C. Illustrative Governance Example

(The following is an illustrative example, not a formal proposition.)

Consider a governance protocol in which voting influence is proportional to delegated stake, while proposal submission is rate-limited per identity.

An adversary controlling a fixed stake allocation may split that allocation across many accounts while preserving total voting weight. Although aggregate stake remains unchanged, the adversary may increase proposal bandwidth, expand committee presence, or amplify visibility in deliberative processes.

If participation limits are enforced only at the account level, the system remains vulnerable to identity amortization. By contrast, if participation is bound to non-transferable execution channels (e.g., device-attested participation), the throughput-bounded component enforces

$$C(s, T) = \Omega(sT)$$

by Theorem 2 and Lemma 5.

This distinction highlights that identity-level security depends not only on protocol rules, but on whether participation constraints are structurally bound to independent channels.

D. Governance Attack Surface

Identity-sensitive governance mechanisms may expose attack surfaces not captured by aggregate-weight security analyses. An adversary distributing a fixed aggregate resource across many identities can potentially obtain disproportionate influence over proposal admission, committee sampling, visibility

amplification, or other participation mechanisms that operate at the identity layer rather than the aggregate resource layer.

These effects arise even when total resource ownership remains unchanged.

E. Mitigation Principle

Within the present framework, preventing identity amortization requires introducing at least one participation constraint that is throughput-bounded and non-transferable. Examples include device-attested execution, actor-bound participation channels, or interaction mechanisms requiring independent execution capacity.

More generally, the results suggest that identity-level security should be treated as a joint property of protocol rules and the structural characteristics of the underlying security resource.

X. LIMITATIONS AND OPEN PROBLEMS

The structural separation established in this work characterizes adversarial identity scaling within the present abstraction. Several limitations and open questions remain.

A. Enforcement of Throughput Constraints

The framework assumes the existence of a per-identity throughput bound τ and activation threshold $r_{\min}$. In practice, enforcing such constraints may be difficult, particularly in environments where participation can be replicated through automation, infrastructure scaling, or outsourced execution.

Verification mechanisms such as device attestation, interaction validation, or behavioral verification may introduce additional latency, coordination complexity, or deployment cost. These costs are intentionally excluded from the model and should be incorporated in deployment-specific analyses.

B. Hybrid and Evolving Resource Classes

Many deployed systems combine multiple resource classes, including capital, computational effort, and human or device-bound participation. While the framework cleanly separates idealized resource types, hybrid constructions may exhibit intermediate scaling behavior.

Lemma 5 establishes partial results for additive hybrid compositions in which throughput-bounded components dominate asymptotically. More general hybrid interactions and non-additive resource compositions remain open theoretical questions.

The structural properties of resources may also evolve over time. Advances in automation, AI-assisted participation, or hardware acceleration may reduce effective non-transferability or increase achievable throughput, potentially narrowing the practical distinction between the two scaling regimes.

C. Economic Coordination and Identity Markets

The present model treats adversarial participation as a resource-allocation problem and abstracts away broader economic coordination dynamics. In practice, delegated participation markets, identity outsourcing, and secondary markets for rate-limited credentials may alter the effective cost structure in ways not captured by the abstraction.

Similarly, governance incentives, token economics, and adaptive protocol policies may influence sustained identity cost over time. Analyzing equilibrium behavior under such adaptive economic conditions remains outside the scope of the present work.

D. Window Granularity and Temporal Modeling

The model partitions time into discrete windows and defines adversarial cost over a horizon of T windows. Consequently, the interpretation of $r_{\min}$ and τ depends on the chosen temporal granularity.

Smaller windows increase renewal frequency under throughput-bounded participation, while larger windows may weaken window-local constraints by allowing resource reuse across longer intervals. Understanding how the scaling bounds depend on temporal granularity—and developing window-independent formulations of the separation results—remains an open direction for future work.

E. Scope of the Framework

The framework is intentionally structural and protocol-agnostic. It is not intended as a complete empirical taxonomy of deployed systems, nor as a universally complete characterization of all Sybil-resistance mechanisms.

Rather, the results isolate resource properties that govern asymptotic adversarial scaling under explicit assumptions, providing a complementary perspective alongside deployment-specific security analyses and protocol-level evaluations.

XI. ILLUSTRATIVE SCALING EXAMPLES

Note. The figures and tables in this section are direct closed-form visualizations of the analytical cost functions derived in Theorems 1–2. They are not empirical evaluations, simulations, benchmarks, or validations of deployed systems. Their purpose is to make the asymptotic scaling regimes concrete across representative parameter ranges.

We plot $C_{\text{par}}(s, T) = s r_{\min} + h(s, T)$ and $C_{\text{bnd}}(s, T) = s T r_{\min}$, with $r_{\min} = 1.0$.

A. Example 1: Identity Scaling

Fixing $T = 100$, the normalized divisible cost $C_{\text{par}}/(sT)$ decays monotonically while the throughput-bounded ratio is constant.

TABLE IV
EXAMPLE 1: NORMALIZED ADVERSARIAL COST $C(s, T)/sT$ ($T = 100$, $r_{\min} = 1.0$, $h = s + T$).

s	C_{par}	C_{par}/sT	C_{bnd}	C_{bnd}/sT
1	102.0	1.0200	100.0	1.0000
5	110.0	0.2200	500.0	1.0000
10	120.0	0.1200	1000.0	1.0000
25	150.0	0.0600	2500.0	1.0000
50	200.0	0.0400	5000.0	1.0000
100	300.0	0.0300	10000.0	1.0000

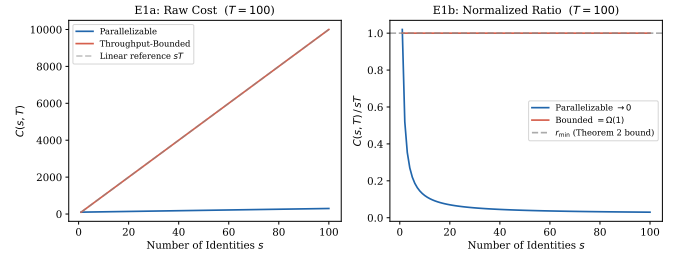


Fig. 3. Example 1: Normalized adversarial cost $C(s, T)/sT$ vs. s ($T = 100$). The divisible-resource ratio decays toward zero, while the throughput-bounded ratio remains fixed at $r_{\min}$.

TABLE V
EXAMPLE 2: NORMALIZED COST $C(s, T)/sT$ ($s = 10$, $r_{\min} = 1.0$, $h = s + T$).

T	C_{par}	C_{par}/sT	C_{bnd}	C_{bnd}/sT
10	30.0	0.3000	100.0	1.0000
25	45.0	0.1800	250.0	1.0000
50	70.0	0.1400	500.0	1.0000
100	120.0	0.1200	1000.0	1.0000
150	170.0	0.1133	1500.0	1.0000
200	220.0	0.1100	2000.0	1.0000

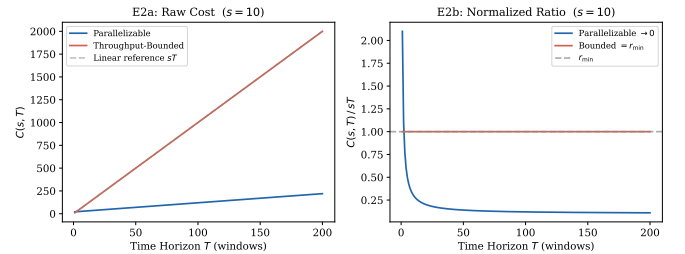


Fig. 4. Example 2: Total adversarial cost $C(s, T)$ vs. T ($s = 10$). The throughput-bounded regime scales linearly with the time horizon, whereas the divisible-resource regime remains asymptotically sublinear relative to sT .

B. Example 2: Time-Horizon Scaling

Fixing $s = 10$, the divisible cost grows as $O(s + T)$ while the throughput-bounded cost scales linearly; the normalized divisible ratio decreases as T grows.

The illustrations above are derived directly from the analytical model and are intended to visualize the asymptotic behavior of the two scaling regimes across representative parameter ranges.

XII. CONCLUSION

This paper proposed an axiomatic framework for analyzing adversarial identity scaling through the structural properties of security resources. We formalized divisibility, temporal reusability, identity transferability, and throughput boundedness, and showed that these properties determine the asymptotic scaling behavior of sustained adversarial cost within the considered model.

Parallelizable resources admit identity amortization, yielding sublinear scaling $C(s, T) = o(sT)$, whereas throughput-bounded, non-transferable resources enforce the linear regime $C(s, T) = \Omega(sT)$. For hybrid systems, Lemma 5 shows that a throughput-bounded component preserves the linear lower bound under additive decomposition.

Illustrative scaling examples and lightweight systems anchoring showed that the structural conditions modeled here correspond to observable participation patterns in deployed PoW, PoS, and governance systems.

The results suggest that aggregate scarcity alone may be insufficient to enforce linear per-identity cost when the underlying resource remains fully divisible and reusable. Identity-level security therefore depends not only on aggregate resource ownership, but also on the structural constraints of the participation resource itself.

Future work includes hybrid resource characterization beyond additive decomposition, practical throughput-enforcement mechanisms, equilibrium analysis under adaptive incentives, and empirical measurement of identity-scaling behavior in deployed systems.

APPENDIX A

TIGHTNESS OF THE SCALING BOUNDS

Both scaling regimes are asymptotically tight.

a) Parallelizable resources.: An adversary acquires $r = sr_{\min}$ once; by temporal reusability this suffices for all T windows. Total cost satisfies $C(s, T) \leq sr_{\min} + h(s, T)$. Since $h(s, T) = o(sT)$ (Assumption 1), $C(s, T)/sT \leq r_{\min}/T + h(s, T)/(sT) \rightarrow 0$ as $s, T \rightarrow \infty$, confirming $C(s, T) = o(sT)$.

b) Throughput-bounded resources.: Setting $r_v(t) = r_{\min}$ for all v, t achieves $C(s, T) = sTr_{\min} = \Theta(sT)$, matching the lower bound of Theorem 2 exactly. Both regimes are therefore tight.

APPENDIX B

ROBUSTNESS UNDER EXTENDED ADVERSARIAL MODELS

The results of Theorems 1–2 are robust under several natural extensions. Under *adaptive identity creation*, divisible resources still permit repartitioning; each newly activated throughput-bounded identity still requires per-window renewal, preserving $C(s, T) = \Omega(sT)$. Under *coalitions*, sustaining s identities still requires s independent channels regardless of coordination structure. Under *stochastic throughput* with $\mathbb{E}[\tau_v(t)] \leq \tau$, expected cost satisfies $\mathbb{E}[C(s, T)] = \Omega(sT)$, with linear scaling holding with high probability under standard concentration bounds. If a system only partially satisfies either regime’s structural properties, cost may interpolate between the two scaling classes.

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